

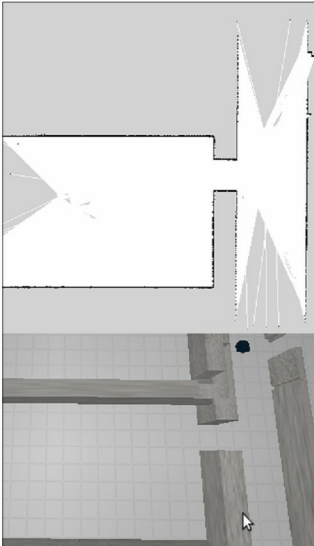


Figure 8: SLAM using gmapping in gazebo with the erratic robot

of the most entertaining aspects of mobile robotics. The philosophy of SLAM is that while navigating, the mobile robot will be able to form a map of its local environment, and thus, can try to answer the question, “Where am I ?” Clearly, this is a *chicken and egg problem*. SLAM is central to the notion of mobile robots surveying unknown terrains probably in some far-off planet. This effort demands sophisticated algorithms and subtle application of the Kalman filter or Rao-Blackwell methods. Interested readers can refer to <http://openslam.org/>, which has an assortment of such algorithms.

SLAM can be realised in ROS using the *gmapping* node. While a simulation similar to that of the erratic robot is running, start *gmapping* in another terminal window using the following command:

```
roslaunch gmapping slam_gmapping scan:=base_scan
```

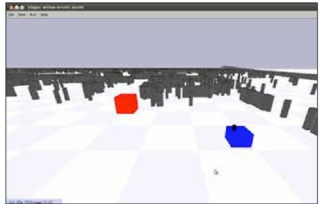


Figure 9: Perspective view of a stage simulation—the blue cube is the erratic robot model

To save the map obtained by SLAM, use the *map_server* command:

```
roslaunch map_server map_saver
```

The map will start to form as the robot is made to go around the environment using teleoperation and it will be saved as a *.pgm* file (Figure 6).

A GUI for ROS

A number of terminal windows are needed to make the best use of ROS. Clearly, a GUI may be a better solution, such as the one being developed as *rxDeveloper* (<http://code.google.com/p/rxdeveloper-ros-pkg/>). This tool provides a pictorial representation of the processes and nodes. It employs launch scripts and ROS tools such as *rviz* and *rxgraph*, which are available as a drop-down menu (Figure 7).

Keep watching this space for more on open source robotics software! **END** 

References

- [1] M Quigley, B Gerkey, K Conley, J Faust, T Foote, J Leibs, E Berger, R Wheeler, and A Y Ng, ‘ROS: An Open Source Robot Operating System’, a paper presented at the Open Source Software workshop of the International Conference on Robotics and Automation (ICRA), 2009.
- [2] JJ Leonard and HF Durrant-Whyte, ‘Simultaneous map building and localisation for an autonomous mobile robot’, a paper presented at Intelligent Robots and Systems (IROS), 1991.

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